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Stated or Sampled? Post-Training Packs a Language Model's Differential Diagnosis into a Single Call

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Abstract

A language model's differential diagnosis can be read from the list it states in one answer or from the spread of its answers over repeated samples, the readout behind pass@k, self-consistency and sampling-based uncertainty. We measure how post-training changes both readouts on DDXPlus, whose synthetic patients carry a rule-based reference differential, with base/post-trained pairs from three open families at 7–32B (Qwen2.5 and Qwen3, every released stage of OLMo 3 7B and OLMo 3.1 32B, and Mistral-7B), zero- and few-shot prompts, and real rare-disease cases. Post-training largely removes the variation between calls. Sampled answers collapse (1.3–2.4 distinct top answers per 32 samples at T=0.7, against 3.8–4.9), and once every model sees worked examples the post-trained models lose 14–30 points of pass@32. One stated list moves closer to the reference, but only by naming conditions with more reference mass, while its probabilities among those conditions move away, and only by about as much as averaging five base-model lists; the gain is robust for Qwen, weak for OLMo 3.1 32B and Mistral, and absent for OLMo 3 7B. Post-training also shifts which severe conditions are named, in both directions, rather than dropping them overall. Majority voting stops helping and agreement between samples weakens as an error signal; averaging the ranks of agreement and stated probability, with no labels, recovers most of what a labeled combination gains. Qwen3-8B's thinking mode reverses much of this: it beats its base model at pass@32 and restores agreement as the better error signal, though its answers stay narrow. Studies of diagnostic breadth should report which readout they use and whether the model reasons first.

Data and Code Availability. All data are public: DDXPlus (Fansi Tchango et al., 2022b) (synthetic patients) and Phenopacket Store v0.1.27 (Danis et al., 2025) (cases from published literature). All models are public checkpoints (Table 4). Prompts and an example vignette are in Appendix A; code and sampled outputs will be released with the camera-ready version.

Institutional Review Board (IRB). This work uses only public synthetic or previously published, de-identified data and involves no human subjects, so it did not require IRB review.

1. Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. A language model's differential can be read in two ways: from the list it states in one answer, or from the distribution of its answers over repeated samples. Much current practice uses the second. Pass@k, self-consistency voting (Wang et al., 2023) and sampling-based uncertainty such as semantic

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entropy (Farquhar et al., 2024) draw on repeated samples, and so does the evidence that reinforcement learning sharpens models, which win at pass@1 while their base models solve more problems given many samples (Yue et al., 2025; Wu et al., 2025; Oh et al., 2026). The two readouts agree for a model that samples from the distribution it states. Instruction-tuned models do not: they verbalize distributions they fail to sample from (Meister et al., 2025; Zhang et al., 2026; Jang et al., 2026), and one aligned answer can cover what many base-model samples spread out (Lake et al., 2025). Jang et al. (2026) show this on synthetic and survey targets, including for Qwen2.5-7B and Mistral-7B, two of our pairs. We ask what it means for differential diagnosis, where every case has a reference differential and a true diagnosis, and what it does to the tools that read the differential from samples. This is a replication and extension in a clinical setting, not a new phenomenon.

The design holds the pretrained weights fixed and varies only the post-training: base/instruct pairs for Qwen2.5-7B, Qwen3-8B, Qwen2.5-32B and Mistral-7B; every released stage (SFT, DPO, RL) of OLMo 3 7B (Team Olmo et al., 2025) and OLMo 3.1 32B, plus a 7B checkpoint trained with RL directly from the base; three medical models fine-tuned from Qwen2.5-7B-Instruct; and Qwen3-8B with its thinking mode on. Each final base/post-trained pair is also run with worked examples in the prompt, so that base models answer in the requested format. Our measurements:

- **Sampled answers collapse** in every family, more so once base models are prompted with worked examples, and the stage that narrows them most depends on the recipe (§4.1).
- **One post-trained call does the averaging.** The stated list improves by naming better conditions, not by stating better probabilities, and by about as much as averaging five base-model lists; severe conditions are reshuffled rather than dropped (§4.2).
- **Sampling-based methods lose value.** Voting stops helping and agreement weakens as an error signal; a label-free rank average of agreement and stated probability is a sound default (§4.3).
- **Reasoning reverses much of it.** Qwen3-8B's thinking mode restores pass@32 and agreement, though not breadth (§4.1, 4.3); on real rare-disease cases the direction replicates (§4.4).

2. Related work

Post-training and the sampled distribution. RL with verifiable rewards improves pass@1 while base models win at large k on math and code (Yue et al., 2025), reweighting solutions already in the base model's support (Wu et al., 2025); a "sharpening tax" appears across 14 base/post-trained pairs on agentic tasks (Oh et al., 2026), though RL can expand the boundary in domains under-represented in pretraining (Cheng et al., 2025). In medicine, the one pass@k analysis we know of studies vision-language models (Jeddi et al., 2026). Instruction tuning and RLHF reduce output diversity (Kirk et al., 2024; West and Potts, 2025; Jiang et al., 2025; Murthy et al., 2025), more so with scale for SFT (Springer et al.,

2026); Liu (2026) place the collapse at DPO in a Zephyr ablation, while Jang et al. (2026) find it growing at each OLMo-2 stage.

Stated versus sampled distributions. Models asked to verbalize a distribution match it better than their samples do, on opinion surveys (Meister et al., 2025) and on synthetic and survey targets (Jang et al., 2026), and they struggle to draw from target distributions directly (Gu et al., 2026). Verbalized sampling keeps most of the base model's diversity through Tülu 3's SFT, DPO and RLVR checkpoints while direct sampling collapses (Zhang et al., 2026), and aligned answers aggregate content that base models spread across samples (Lake et al., 2025). Our stated readout is the verbalized top-k elicitation of Tian et al. (2023). For error detection, sample consistency usually beats verbalized confidence and hybrids beat both (Xiong et al., 2024; Lyu et al., 2024), including for medical diagnosis (Savage et al., 2025; Ren et al., 2026), and aligned models' samples can collapse enough to leave sampling-based uncertainty without signal (Liu, 2026). We add a clinical task with a reference differential and a true diagnosis, three pipelines at 7–32B with a few-shot control, and the consequences for aggregation and uncertainty.

Medical models and differential diagnosis. LLMs have been evaluated as differential-diagnosis aids on clinicopathological cases (Kanjee et al., 2023; McDuff et al., 2025); on Phenopacket Store, LLMs trail the phenotype-driven tool Exomiser (Reese et al., 2026). Medical post-training recipes include verifier-guided SFT and PPO (Chen et al., 2025), rule-based RL (Liu et al., 2025) and reasoning distillation (Huang et al., 2026); RL and reward models tend to favor confident answers (Leng et al., 2025; Damani et al., 2026; Mei et al., 2026). DDXPlus was built for diagnosis systems trained on differentials (Fansi Tchango et al., 2022a) and scores predicted against reference differentials as sets; H-DDx proposes hierarchical F1 over ICD-10 (Lim et al., 2025), and MEDDxAgent studies interactive diagnosis on it (Rose et al., 2025). Puri et al. (2026) show on DDXPlus with Qwen3-8B that single-answer RL collapses sampled answers and train models to output several answers per completion. We compare stated and sampled differentials against the reference *distribution* across public post-training pipelines.

3. Setup

Tasks. **DDXPlus** (Fansi Tchango et al., 2022b): synthetic patients from a rule-based medical knowledge base, each with a true pathology and a reference differential, the engine's posterior over 49 conditions. We use 980 test patients (20 per condition) rendered by template as questionnaire-style vignettes without exam or test results (example in Appendix A). The prompt lists the 49 conditions and asks for the five most likely, with probabilities, as JSON, so this is a closed-set task. **Phenopacket Store** (Danis et al., 2025): 1,401 published rare-disease cases covering 749 diseases (at most two per disease), rendered as phenotype vignettes with an open-ended differential; names are graded against the case's diagnosis via Mondo as exact, related (a specific ancestor group or subtype) or wrong (Appendix A).

Models and sampling. Qwen2.5-7B, Qwen3-8B, Qwen2.5-32B and Mistral-7B-v0.3, each with its official post-trained version; OLMo-3-1025-7B and OLMo-3-1125-32B with the SFT,

DPO and final RLVR (Instruct) checkpoints of OLMo 3 7B and OLMo 3.1 32B, plus OLMo-3-7B-RL-Zero-General, trained with RL from the 7B base; and AlphaMed (rule-based RL), HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation), all fine-tuned from Qwen2.5-7B-Instruct (Table 4). We draw 32 samples per case with vLLM at T=0.7 (top-p 0.95) and, for the 7–8B models, T=1 (top-p 1), each model with its own chat template (base models without one get a post-trained sibling's) and Qwen3's thinking mode off unless stated. A sample's *top answer* is valid if it is one of the 49 conditions and its *list* if it names at least one; invalid answers count as wrong. With this zero-shot prompt, base models are often invalid (up to 47% of samples for OLMo and 52% for Mistral), so every final pair is re-run with three worked examples from the training split (Appendix D.3), on which every model is valid in over 99% of samples.

Two readouts. The *sampled* readout is the distribution of top answers over a case's samples: pass@k (Chen et al., 2021), the number of distinct top answers, the reference mass they cover (coverage@32), and total-variation distance (TV) of their empirical distribution to the reference. The *stated* readout is one sample's list, with its stated probabilities renormalized over the conditions it names (uniform if missing), averaged per case over valid samples: its TV to the reference, also after removing the true pathology from both, DDXPlus's set F1 against the reference's support (DDF1), the reference mass it names, and whether it names each severe condition (DDXPlus severity 1–2, the dataset's label) that the reference weights at least 0.1. A list q with named set L has TV $= (1-\sum_{c\in L} p_c)+\sum_{c\in L}\max(0, p_c-q_c)$ against reference p: the mass it leaves out plus the mass it *misallocates* among the conditions it names. The *single-call answer* is the top answer of a model's first valid sample.

Statistics. DDXPlus has 20 cases per true pathology, so intervals are 95% percentile intervals from a bootstrap that resamples the 49 pathologies with all their cases (10,000 resamples) unless stated; p-values use the add-one correction, and the headline base-vs-post differences are Holm-corrected over 112 tests (Appendix G). Appendix intervals are from a paired case bootstrap, two to three times narrower, unless marked as clustered. AUROCs use average ranks for ties. With five to six pipelines, results describe these checkpoints rather than post-training in general.

4. Results

4.1. Post-training collapses the sampled answers

Table 1 compares each base model with its post-trained versions. At 7–8B the post-trained Qwen model is far better on its first answer (+12.6 to +27.5 points), but its pass@k curve crosses the base model's at k≈4–8, and at k=32 the base model leads by 9.7–14.6 points (Figure 2 in Appendix C.1). At 32B the zero-shot losses are smaller and uncertain (Qwen2.5-32B −4.9 $[-10.2, +0.6]$, OLMo 3.1 32B −4.0 $[-6.8, -1.2]$, neither surviving correction). With worked examples in every prompt, base models answer validly and post-training costs 14–30 points of pass@32 in all six final pairs, including both 32B pairs and Mistral-7B (Appendices D.3 and D.4). Post-trained models give 1.3–2.4 distinct top answers per 32 samples at T=0.7, against 3.8–4.9 for their base models. Coverage@32 rewards spreading answers: a guesser drawing uniformly from the 49 conditions covers 0.48 and one drawing from the average reference differential 0.56, more than any model. TV does not reward

spreading for its own sake: a sampler that always gives each case's reference mode scores 0.694, and one that always gives the true pathology 0.714, both better than the patient-blind guesser (0.774; Table 2). Base models' samples lie in between (0.73–0.76 at T=0.7), and post-trained samples are farther than either (0.76–0.85): they are concentrated, and often on the wrong condition. Which stage narrows the samples most depends on the recipe. In OLMo 3 7B, distinct answers fall from 4.93 to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR, and the RLVR step alone costs 19.1 [13.8, 24.6] points of pass@32 against the DPO model; in OLMo 3.1 32B, SFT costs most (12.1 [6.8, 18.0]), DPO recovers most of it, and RLVR costs 2.8 [0.5, 5.3] (Appendix C.5). RL-Zero, trained with RL from the 7B base, loses 13.3 [7.3, 19.7] points at T=0.7, about as much as SFT, and none detectably at T=1 (-1.8 [-5.8, $+2.1$]; Appendix C.4). Qwen3-8B with its thinking mode on is the exception: the same weights beat the base model at pass@32 (0.815 against 0.732; +8.4 [1.7, 15.3], not surviving correction) with samples as close to the reference as the base model's (TV 0.745 against 0.750), though they stay narrow (1.76 distinct answers; Appendix D.5). Severe conditions also become rarer among sampled top answers, but even base models consider them less often than the patient-blind guesser (Appendix C.2), so we assess them on the stated list.

4.2. One post-trained call does the averaging

For every Qwen model and OLMo 3.1 32B, the post-trained model's single list is closer to the reference than its base model's (TV -0.043 to -0.088; Table 2, Figure 1a), also with the true pathology removed from both (-0.018 to -0.056; Qwen3-8B at T=0.7 does not survive correction) and by DDXPlus's set F1 (+0.040 to +0.083). OLMo 3 7B's Instruct lists are not closer, and without the true pathology they are farther (+0.034 at T=0.7). With worked examples in every prompt the Qwen gains persist or grow, OLMo 3.1 32B keeps only a higher reference mass, and Mistral-7B's gain is the smallest (TV -0.036; Appendices D.3, D.4).

Two analyses show what this gain is. First, it lies entirely in the choice of conditions. The reference mass a list leaves out falls, while the mass it misallocates among the conditions it names rises after post-training in seven of eight settings (+0.011 to +0.033; Table 2) and in every few-shot pair. Second, averaging calls closes most of the gap: the mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700, Qwen2.5-7B 0.717 against 0.717, OLMo 3.1 32B 0.691 against 0.701; diamonds in Figure 1a). With five lists on both sides, post-training is still closer for Qwen2.5 (-0.024 to -0.035) but not detectably for Qwen3-8B, OLMo 3.1 32B or Mistral, and farther for OLMo 3 7B (Appendix D.6). Direct-answer post-training thus mostly packs into one call what the base model spreads across calls, as Lake et al. (2025) find for opinions. Qwen3-8B's thinking mode goes further: one of its lists (TV 0.670) beats five averaged base lists, and five of its lists beat five base lists by 0.049.

Severe conditions. Post-trained lists name the severe conditions that the reference weights at least 0.1 more often in every setting (Qwen +9.0 to +13.8 points, OLMo 3 7B +6.0 and +8.7; OLMo 3.1 32B +2.2, n.s.), yet one list still names only 0.32–0.47 of them, against 0.975 for the reference's own top five. Condition by condition (any reference weight; Table 11), post-training reshuffles rather than drops: after false-discovery-rate correction,

Comparison	Model	Sampled top answers pass@1	pass@32	Distinct	Cov@32	Drawn from stated lists pass@32	Distinct
Qwen3-8B, T=0.7	base	0.388	**0.730**	**3.76**	**0.322**	0.780	7.79
	post-trained	**0.558**	0.607	1.27	0.217	**0.811**	4.96
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**0.407**	**0.832**	9.82
	post-trained	**0.556**	0.643	1.53	0.236	0.820	5.40
Qwen2.5-7B, T=0.7	base	0.352	**0.695**	**4.91**	**0.335**	0.772	10.68
	instruct	**0.478**	0.598	1.82	0.239	**0.802**	6.81
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**0.422**	0.785	12.44
	instruct	**0.464**	0.663	2.65	0.290	**0.827**	7.76
Qwen2.5-32B, T=0.7	base	0.500	**0.795**	**3.78**	**0.352**	0.850	8.79
	instruct	**0.604**	0.746	1.68	0.276	**0.880**	6.49
OLMo 3 7B, T=0.7	base	0.178	**0.705**	**4.93**	**0.351**	**0.719**	8.66
	SFT	0.222	0.559	4.01	0.257	0.618	7.18
	SFT+DPO	**0.324**	0.613	3.30	0.272	0.702	6.88
	SFT+DPO+RLVR	0.266	0.422	2.37	0.195	0.577	5.65
	RL-Zero (from base)	0.295	0.572	4.06	0.286	0.701	9.66
OLMo 3.1 32B, T=0.7	base	0.326	**0.796**	**4.32**	**0.347**	0.839	8.11
	SFT	0.461	0.674	2.94	0.265	0.759	7.94
	SFT+DPO	**0.584**	0.784	2.29	0.293	**0.868**	6.64
	SFT+DPO+RLVR	0.560	0.756	2.15	0.283	0.853	6.75
Chance	uniform guesser	0.020	0.48	23.7	0.48		
	reference-prior guesser	—	0.44	21.9	0.56		

Table 1: Sampled readout on DDXPlus (980 cases, 32 samples). Distinct: distinct top answers per case; Cov@32: reference mass they cover. *Drawn from stated lists*: each sample's top answer replaced by a draw from its own stated list. Bold: better model of each pair (OLMo: of the block). Invalid answers count as wrong, which mostly penalizes base models. Chance: guessers that ignore the patient, drawing uniformly from the 49 conditions or from the reference differential averaged over test cases.

2–7 of the 17 severe conditions are named significantly less often and 0–6 significantly more often, and the number named less often is never larger than chance would give. The shifts have a pattern. The Qwen models name acute coronary syndromes and pulmonary embolism more often and stable angina or acute pulmonary edema less often, OLMo 3 7B names pulmonary embolism, epiglottitis and myocarditis more often and the coronary syndromes less often, and OLMo 3.1 32B names acute pulmonary edema and NSTEMI less often with no significant gains. These vignettes lack the ECG and biomarkers that separate the coronary syndromes, so the shifts track the reference engine, not clinical correctness.

4.3. Consequences for sampling-based methods

Aggregation. Majority voting over 32 samples (self-consistency) raises base models' accuracy by 3.4–8.4 points at T=0.7 but adds at most 1.9 for the final post-trained models (Qwen2.5-32B-Instruct, $[0.8, 3.1]$; the others n.s.; Table 3). Models that reason before answering still gain (HuatuoGPT-o1 +4.8, m1 +3.8, Qwen3-8B with a CoT prompt +4.5), and Qwen3-8B with thinking on gains +1.5 (n.s.). For a top-three differential, the top three of one list contains the true pathology more often than the three most frequent of 32

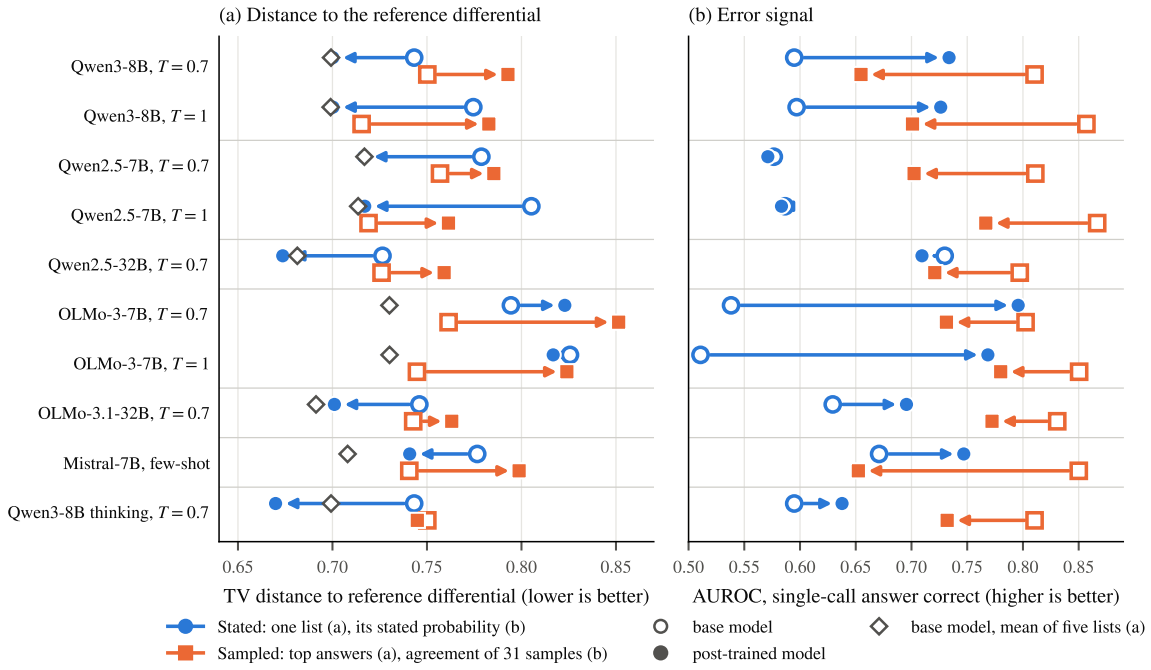


Figure 1: Base (hollow) → post-trained (filled) on DDXPlus (980 cases; OLMo: base → final Instruct; Mistral-7B with the few-shot prompt; the last row compares Qwen3-8B-Base with Qwen3-8B with thinking on). (a) TV to the reference differential of one stated list (blue circles) and of the sampled top answers (orange squares); grey diamonds: the mean of the base model's first five lists. Post-training moves the list toward the reference and the samples away from it, but five averaged base lists get about as close as one post-trained list, except with thinking on. (b) AUROC for the single-call answer being correct, of its stated probability (blue) and of agreement across 31 further samples (orange).

sampled answers for the post-trained Qwen models (+5.6 to +13.7 points), while for every base model the samples win by 7.3–13.6.

Uncertainty. We score each signal by how well it ranks the correctness of the single-call answer (AUROC). The stated probability needs no extra calls; *agreement*, the share of other samples with the same top answer, needs 4 or 31. For base models, agreement across 31 samples beats the stated probability by 0.07–0.26. Post-training weakens agreement in every family (Qwen3-8B 0.810 → 0.655, Qwen2.5-7B 0.811 → 0.702, Qwen2.5-32B 0.797 → 0.721, OLMo 3 7B 0.802 → 0.731, OLMo 3.1 32B 0.831 → 0.772; Figure 1b), significantly after correction for Qwen but not for OLMo, consistent with Liu (2026). The stated probability then wins for Qwen3-8B (+0.079 $[0.011, 0.143]$ over 32-sample agreement) and, against five-sample agreement, for OLMo-3-7B-Instruct (+0.128 $[0.052, 0.203]$); for the other final models agreement stays better or ties, in line with Savage et al. (2025). Thinking returns Qwen3-8B to the base pattern (agreement 0.732 against 0.638).

Comparison (T=0.7)	Model	TV to reference↓ One list	Samples	List w/o gold	Reference mass named	Misallocated↓	DDF1	Severe named
Qwen3-8B	base	0.743	**0.750**	0.833	0.334	**0.077**	0.276	0.347
	post-trained	**0.700**	0.793	**0.815**	**0.396**	0.096	**0.318**	**0.437**
Qwen2.5-7B	base	0.779	**0.757**	0.856	0.291	**0.070**	0.245	0.345
	instruct	**0.717**	0.785	**0.816**	**0.364**	0.081	**0.307**	**0.439**
Qwen2.5-32B	base	0.726	**0.726**	0.830	0.351	**0.078**	0.282	0.382
	instruct	**0.674**	0.759	**0.787**	**0.418**	0.092	**0.333**	**0.472**
OLMo 3 7B	base	**0.794**	**0.761**	**0.850**	**0.267**	**0.062**	**0.225**	0.256
	Instruct (RLVR)	0.823	0.851	0.884	0.257	0.080	0.220	**0.317**
OLMo 3.1 32B	base	0.746	**0.743**	0.835	0.334	**0.080**	0.273	0.370
	Instruct (RLVR)	**0.701**	0.763	**0.812**	**0.391**	0.092	**0.312**	**0.392**
Chance	patient-blind	0.794	0.774	—	0.238	0.032	0.272	0.149
Oracle	reference	0.265	0.18–0.25	—	0.735	0.000	—	0.975
	one answer: reference mode	—	0.694	—	—	—	—	—
	one answer: true pathology	—	0.714	—	—	—	—	—

Table 2: Stated vs. sampled readouts on DDXPlus (T=0.7, 980 cases, valid samples). *One list*: one sample's stated differential (per-case mean over valid samples). *Samples*: empirical distribution of top answers. *List w/o gold*: TV after removing the true pathology from list and reference. A list's TV is exactly the reference mass it leaves out (1− mass named) plus the mass it *misallocates* among the conditions it names. DDF1: set F1 of the named conditions against the reference's support. *Severe named*: share of severe conditions with reference probability ≥ 0.1 that one list names. Chance: the top five of the average reference differential as a fixed list, and 32 draws from that average as samples. Oracle: the reference's own top five as the list and as many reference draws as valid samples; and samplers that always give one answer, each case's reference mode or its true pathology. Bold: better model of each pair.

Which signal to use. Across 56 model–setting combinations (descriptive, since they share weights; Appendix D.7), the stated probability's advantage over five-sample agreement tracks how often five samples all agree (Spearman 0.75), partly by construction, since agreement is then tied. The practical default needs no labels: averaging the two signals' ranks gives a held-out AUROC of 0.746 for the 39 post-trained combinations, against 0.695 for always using agreement and 0.671 for always using the stated probability. A logistic combination fitted on 100 labeled cases adds little (0.753), and picking the better single signal on those cases does worse (0.720).

Ambiguity. For every model and signal, the Spearman correlation between uncertainty and the entropy of the case's reference differential lies between −0.11 and +0.16, except AlphaMed's stated probability (+0.30): by either readout, a model's uncertainty says almost nothing about how ambiguous a case is.

4.4. Real rare-disease cases and medical post-training

Real rare-disease cases. Phenopacket Store has no reference differential, and 7–8B models are near the floor (pass@32 2.5–7.0%), so we count an answer as correct if it is the diagnosis or a closely related Mondo class (36–130 correct single-call answers per model; Appendix E.3) and read these results as directional. Majority voting helps the base models

Model	Single-call accuracy	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct		
				Stated p	Agree@5	Agree@32
Qwen3-8B-Base	0.456	+3.4	−7.3	0.595	0.772	**0.810**
Qwen3-8B	0.556	+0.3	+13.7	**0.734**	0.613	0.655
Qwen3-8B (CoT prompt)	0.648	+4.5	+0.0	0.718	0.725	**0.778**
Qwen3-8B (thinking)	0.672	+1.5	+3.0	0.638	0.693	**0.732**
Qwen2.5-7B	0.371	+5.2	−8.9	0.576	0.764	**0.811**
Qwen2.5-7B-Instruct	0.480	+0.8	+7.3	0.571	0.653	**0.702**
Qwen2.5-32B	0.510	+7.0	−8.3	0.730	0.759	**0.797**
Qwen2.5-32B-Instruct	0.598	+1.9	+5.6	0.709	0.663	**0.721**
OLMo-3 7B base	0.291	+4.0	−13.6	0.538	0.761	**0.802**
OLMo-3-7B-Instruct	0.267	−0.2	+1.9	**0.796**	0.668	0.731
OLMo-3.1 32B base	0.441	+8.4	−13.2	0.629	0.805	**0.831**
OLMo-3.1-32B-Instruct	0.562	+0.9	−0.6	0.696	0.712	**0.772**
AlphaMed-7B	0.528	+0.9	+0.6	0.673	0.663	**0.715**
HuatuoGPT-o1-7B	0.532	+4.8	−0.4	0.590	0.715	**0.775**
m1-7B	0.495	+3.8	−4.6	0.680	0.736	**0.782**

Table 3: Aggregation and uncertainty on DDXPlus (T=0.7, 980 cases). Single-call accuracy uses the first valid sample, so it can exceed pass@1, which counts invalid answers as wrong. Vote@32 gain: accuracy of the majority top answer over all valid samples minus single-call accuracy (points). Top-3: rate at which one list's top three contains the true pathology minus the rate for the three most frequent sampled answers (points). AUROC: stated probability of the single-call answer (no extra calls) vs. agreement of 4 or 31 further samples; bold marks the best signal.

by 3.1–4.6 points but the final post-trained models by only 0.4–1.5. Post-training weakens agreement as an error signal (Qwen3-8B 0.830 → 0.763, Qwen2.5-7B 0.845 → 0.745, OLMo 3 7B 0.814 → 0.726) and strengthens the stated probability (0.567 → 0.707, 0.598 → 0.717, 0.464 → 0.652), but unlike on DDXPlus agreement stays the better signal (by 0.03–0.07 after post-training, detectably only for Qwen3-8B), and the three most frequent sampled answers still contain a correct diagnosis more often than one list's top three (Qwen3-8B: a tie).

Medical models. Medical post-training on Qwen2.5-7B-Instruct does not narrow the samples further: AlphaMed (rule-based RL) improves on its parent at large k (+8.4 [2.6, 14.7] points at k=32) with unchanged diversity (1.87 vs. 1.82 distinct answers), and HuatuoGPT-o1 and m1, both trained on reasoning traces, widen the sampled differential (2.49 and 3.29; Appendix E.1). On Phenopacket Store, stated confidence is not near the floor: the RL-trained medical models (AlphaMed; HuatuoGPT-o1 after SFT) raise it from 0.51 for the parent to 0.67–0.69 while accuracy gains stay under 1.5 points, and every OLMo 3 7B stage states 0.73–0.84 at about 1% accuracy (Appendix E.2).

5. Discussion and limitations

Jang et al. (2026) show that instruction-tuned models cannot sample from distributions they can describe. In differential diagnosis, against a reference with a known true pathology, we find the same split and a simpler account of it: direct-answer post-training largely removes the variation between calls. The sampled answers collapse, and the single stated list improves mainly because one call now returns about what several base-model calls give on average, by naming better conditions rather than by stating better probabilities. Evaluations that read the differential from samples (pass@k at large k, diversity, self-consistency, agreement-based uncertainty) therefore report a loss that a clinician reading one answer would not see, and these tools lose value on post-trained models; a label-free rank average of agreement and stated probability is a reasonable default error signal. Conversely, a better stated list is no evidence that post-training adds diagnostic knowledge: averaged base lists match it, every model stays far from the reference, neither readout tracks how ambiguous a case is, and the reshuffling of severe conditions is something a must-not-miss review would need to check model by model. These effects belong to answering directly: Qwen3-8B's thinking mode restores large-k accuracy and agreement as an error signal, though not breadth. Which stage narrows the samples most is a property of the recipe.

Limitations. DDXPlus is synthetic and closed-set: its reference differential is a rule-based engine's posterior over questionnaire answers without exam or test results, its test set is class-balanced, and the prompt lists all 49 conditions, so TV, coverage and severe-condition naming measure agreement with that engine, not clinical correctness, and no clinician reviewed the lists or the references. Severity is the dataset's label, and the prompt asks for the most likely conditions, not for must-not-miss ones. The few-shot control uses one temperature and one set of examples, which show the reference's own lists. We study open 7–32B models from three families, one reasoning mode, one RL-Zero checkpoint and no frontier models, and the 32B models only at T=0.7. The signal combinations are validated only within DDXPlus. Phenopacket accuracy is near the floor, its grader is rule-based and not validated against clinicians, and its cases may be in pretraining data.

Statement on AI Use. An AI assistant (Claude, Anthropic) was used to write and run analysis code, to draft and edit text, and to check citations. The authors designed the study, reviewed all code, results and text, and take responsibility for the content.

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Appendix A. Prompts, vignettes and grading

DDXPlus. Each patient's coded evidence is rendered by a fixed template into a vignette of the patient's answers and relevant history; we fix known mistranslations in the English evidence descriptions. An example (test case with true pathology chronic rhinosinusitis; the reference differential gives viral pharyngitis 0.17, chronic rhinosinusitis 0.16, cluster headache 0.15, acute rhinosinusitis 0.14, bronchitis 0.14, URTI 0.12):

> `A 13-year-old female presents for evaluation. Symptoms (patient answers): Do you have pain somewhere, related to your reason for consulting? Yes. Characterize your pain: burning. Do you feel pain somewhere? mouth, forehead, cheek(R), cheek(L), nose. How intense is the pain? (scale 0-10) 3. […] Have you lost your sense of smell? Yes. Do you have nasal congestion or a clear runny nose? Yes. Do you have a cough? Yes. Relevant history and risk factors: Have you had a cold in the last 2 weeks? Yes. […] Do you have polyps in your nose? Yes. […]`

System prompt: `You are an experienced emergency and primary care physician.` The user turn is the vignette followed by: `Based on the patient's answers above, give your differential diagnosis. Choose ONLY from this list of possible conditions: [the 49 conditions, one per line]. List the 5 most likely conditions from the list, most likely first, with your probability that each is the correct diagnosis. Use the condition names exactly as written in the list. End your answer with a JSON array in exactly this format: [{"diagnosis": "<name>", "probability": <number between 0 and 1>}, ...].` The full-distribution prompt (§D.1) instead asks for a probability on every condition, summing to one, most likely first. The plain and CoT variants are described in §C.3. Base models receive the same chat-formatted prompt; OLMo base models use their SFT checkpoint's chat template.

Phenopacket Store. The vignette lists the case's present and (up to ten) excluded HPO phenotypes with sex and age. System prompt: `You are an expert clinical geneticist and diagnostician.` Instructions: `List the 5 most likely specific diagnoses (use the specific disease or subtype name, e.g. 'Spastic paraplegia 7' rather than 'hereditary spastic paraplegia', when you can), most likely first, with your probability that each is the correct diagnosis`, followed by the same JSON format.

Grading open-ended names. Each predicted name is mapped to Mondo by exact lookup against Mondo labels and synonyms (also trying any parenthesized abbreviation and the name without it), then by fuzzy matching (token-sort ratio ≥ 92). It is *exact* if it maps to the Mondo class of the case's OMIM diagnosis, and *related* if it maps to a specific ancestor group of it (at most two is-a steps up, with at most 150 descendants) or to one of its subtypes; siblings sharing only a broad parent are wrong. The grader passes hand-written checks (e.g. "Hereditary spastic paraplegia type 7 (SPG7)" is exact for SPG7, "Hereditary spastic paraplegia" is related, and Turner syndrome is wrong for Jacobsen syndrome), but we did not validate it against clinician grading, so errors at the scale of the 1–2 point effects we report on this dataset are possible.

Appendix B. Models

Model	Parent	Post-training	Role
Qwen2.5-7B	n/a	none (pretrained)	base
Qwen2.5-7B-Instruct	Qwen2.5-7B	general SFT + RLHF	instruct; medical parent
AlphaMed-7B (Liu et al., 2025)	Qwen2.5-7B-Instruct	rule-based RL	medical
HuatuoGPT-o1-7B (Chen et al., 2025)	Qwen2.5-7B-Instruct	SFT + PPO (verifier)	medical
m1-7B-23K (Huang et al., 2026)	Qwen2.5-7B-Instruct	SFT distillation	medical
Qwen3-8B-Base	n/a	none (pretrained)	base
Qwen3-8B	Qwen3-8B-Base	general post-training	instruct
Qwen2.5-32B	n/a	none (pretrained)	base
Qwen2.5-32B-Instruct	Qwen2.5-32B	general SFT + RLHF	instruct
OLMo-3-1025-7B (Team Olmo et al., 2025)	n/a	none (pretrained)	base
OLMo-3-7B-Instruct-SFT	OLMo-3-1025-7B	SFT	stage 1
OLMo-3-7B-Instruct-DPO	OLMo-3 SFT	DPO	stage 2
OLMo-3-7B-Instruct	OLMo-3 DPO	RLVR	stage 3 (instruct)
OLMo-3-7B-RL-Zero-General	OLMo-3-1025-7B	RL (LM judge)	RL from base
OLMo-3-1125-32B	n/a	none (pretrained)	base
OLMo-3.1-32B-Instruct-SFT	OLMo-3-1125-32B	SFT	stage 1
OLMo-3.1-32B-Instruct-DPO	OLMo-3.1 SFT	DPO	stage 2
OLMo-3.1-32B-Instruct	OLMo-3.1 DPO	RLVR	stage 3 (instruct)

Table 4: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training. OLMo 3 and OLMo 3.1 release every stage of their instruct pipelines.

Appendix C. The sampled differential in detail

C.1. General post-training trades coverage for top-1 accuracy

Table 5 and Figure 2 compare each base model with its post-trained counterpart on DDXPlus. In both families the post-trained model is much better on its first answer: +27.5 points for Qwen3 and +22.5 for Qwen2.5 at T=1 (intervals $[+25.3, +29.6]$ and $[+20.7, +24.4]$). The advantage shrinks with k and reverses: the curves cross at k≈4–8, and at k=32 the base model leads by 14.6 points (Qwen3, $[+12.2, +17.1]$) and 10.3 points (Qwen2.5, $[+7.8, +12.8]$). The pattern is the same at T=0.7 for Qwen3, crossing at k=4 with a base lead of 12.2 points at k=32 ($[+9.9, +14.9]$). Unparsable base-model samples count as wrong,

Pair	Model	pass@1	pass@32	Distinct	Entropy	TV↓	Cov@32	ECE↓
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**1.31**	**0.712**	**0.407**	0.214
	post-trained	**0.556**	0.643	1.53	0.18	0.783	0.236	**0.061**
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**1.58**	**0.715**	**0.422**	0.303
	instruct	**0.464**	0.663	2.65	0.49	0.761	0.290	**0.137**
Qwen3-8B, T=0.7	base	0.388	**0.730**	**3.76**	**0.78**	**0.750**	**0.322**	0.121
	post-trained	**0.558**	0.607	1.27	0.11	0.793	0.217	**0.063**

Table 5: DDXPlus, 32 samples per case (980 cases). The reference differential has mean entropy 1.87. Distinct and entropy refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability.

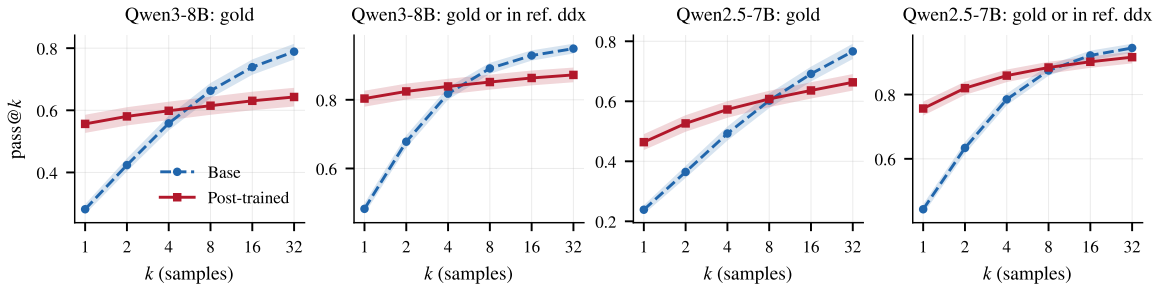


Figure 2: pass@k on DDXPlus at T=1 (980 cases, 32 samples), base model (dashed) vs. its post-trained version (solid), with 95% case-bootstrap bands. "Gold": the top-1 answer is the true pathology; "gold or in ref. ddx": it is the true pathology or appears in the reference differential. In both families the post-trained model wins at small k and loses at large k.

so these gaps are lower bounds. Qwen2.5 at T=0.7 behaves the same way: +12.6 points at k=1 [+10.7, +14.5] and a base lead of 9.7 at k=32 [+7.3, +12.0].

C.2. The sampled differential collapses, while top-1 calibration improves

The gap at large k reflects a collapsed *sampled* differential. Across 32 samples, the post-trained Qwen models name 1.3–2.6 distinct diagnoses per case, against 3.8–8.2 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.58 for base models and 1.87 for the reference differential (Table 5). Sampling at T=1 widens the base models' answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at T=0.7, 0.18 at T=1), so the collapse lives in the model's own distribution rather than in the decoding settings. As a result, post-trained models cover less of the reference differential: 0.24 vs. 0.41 of its probability mass at k=32 for Qwen3, and 0.29 vs. 0.42 for Qwen2.5. Their top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.21 and 0.14 vs. 0.30). On every single-answer criterion (accuracy, small-k pass@k, calibration), the post-trained model therefore looks

strictly better, and the loss of breadth in its samples is invisible. Whether the breadth is lost from the model's stated differential as well is a separate question (§4.2).

Which diagnoses drop out? For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 6), and the severe tier (DDXPlus severity 1–2, the dataset's label rather than a clinical consensus list) is not spared. This metric needs a chance reference, though: a guesser that picks uniformly among the 49 conditions scores about 0.48 on it, and one that draws from the average reference differential 0.56–0.61, above every model, base models included. Considered@32 therefore measures the breadth of the sampled answers, not clinical judgement; we assess can't-miss conditions on the stated list instead (§D.2). For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.51 to 0.34 for Qwen2.5 at T=1. The differences are -18.4 $[-20.8, -15.9]$ and -17.0 $[-19.7, -14.2]$ points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95% for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives, a bias toward the most dangerous candidate rather than a differential.

C.3. Robustness to prompt and output format

Prompt. We repeat the Qwen3-8B pair with two other prompts (Table 7). *Plain* drops the system prompt and rewords the instructions. *CoT* asks the model to reason step by step before answering; Qwen3's thinking mode stays off in all runs. The plain prompt reproduces the pattern: the post-trained model gains $+23.0$ $[+20.9, +25.1]$ points at k=1 and loses -7.1 $[-9.3, -5.0]$ at k=32 (T=0.7; at T=1, $+34.2$ and -8.7 $[-11.0, -6.4]$), with 1.3–1.5 distinct answers against 3.7–5.6. Asking for reasoning changes the picture. With CoT, the post-trained model's pass@32 rises from 0.607 to 0.835: it leads the base model at every k at T=0.7 ($+4.7$ $[+2.9, +6.6]$ at k=32) and ties it at T=1 (0.0 $[-2.1, +1.9]$). Its samples remain far less diverse than the base model's (2.1–2.2 vs. 3.9–6.3 distinct answers), so reasoning recovers coverage at k=32 without restoring sampling diversity. The coverage loss is therefore largest when the model answers directly, as in most clinical decision-support deployments, and reasoning at inference time mitigates it. This matches the medical models: the reasoning-trained descendants widen the differential (§E.1).

Comparison	Considered@32 (base → post-trained) Severe (1–2)	Moderate (3)	Mild (4–5)
Qwen3-8B, T=1	0.45 → 0.27	0.50 → 0.32	0.45 → 0.20
	−18.4 [−20.8, −15.9]	−18.1 [−20.7, −15.7]	−24.2 [−27.0, −21.4]
Qwen2.5-7B, T=1	0.51 → 0.34	0.50 → 0.37	0.38 → 0.23
	−17.0 [−19.7, −14.2]	−12.1 [−14.7, −9.7]	−15.0 [−17.4, −12.6]
Qwen3-8B, T=0.7	0.34 → 0.24	0.44 → 0.29	0.31 → 0.18
	−9.8 [−11.8, −7.9]	−15.8 [−18.2, −13.5]	−13.4 [−15.8, −11.1]
Qwen2.5-7B, T=0.7	0.40 → 0.28	0.41 → 0.32	0.28 → 0.17
	−12.4 [−14.7, −10.2]	−9.1 [−11.2, −6.7]	−10.3 [−12.3, −8.3]
Instruct → AlphaMed, T=0.7	0.28 → 0.32	0.32 → 0.34	0.17 → 0.21
	+4.3 [+2.0, +6.6]	+2.2 [+0.4, +4.0]	+3.9 [+2.4, +5.4]
Instruct → HuatuoGPT-o1, T=0.7	0.28 → 0.36	0.32 → 0.37	0.17 → 0.27
	+8.4 [+6.1, +10.8]	+6.0 [+4.0, +7.9]	+9.5 [+7.5, +11.5]
Instruct → m1, T=0.7	0.28 → 0.41	0.32 → 0.41	0.17 → 0.29
	+13.5 [+10.8, +16.2]	+9.7 [+7.2, +12.2]	+11.4 [+9.2, +13.6]
Base → AlphaMed (base-RL), T=0.7	0.40 → 0.39	0.41 → 0.40	0.28 → 0.27
	−0.7 [−3.0, +1.7]	−0.4 [−2.6, +1.7]	−0.4 [−2.4, +1.7]
Base → `acc` GRPO (8B), T=0.7	0.34 → 0.31	0.44 → 0.42	0.31 → 0.27
	−2.8 [−4.4, −1.3]	−2.1 [−3.7, −0.6]	−4.3 [−6.2, −2.5]
Base → `gm` GRPO (8B), T=0.7	0.34 → 0.32	0.44 → 0.43	0.31 → 0.28
	−1.5 [−3.2, +0.1]	−1.3 [−3.0, +0.3]	−3.0 [−4.8, −1.2]

Table 6: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% case-bootstrap intervals. All rows use the 980 test cases; GRPO rows are seed 0. General post-training lowers consideration in every tier, including the can't-miss tier; medical post-training raises it; our GRPO from the base model loses at most 4 points, and with the group reward the severe-tier loss is not significant.

Output format. Base models more often give invalid answers: unparsable, or with a top answer outside the 49 conditions. This affects 12% of Qwen3-8B-Base samples at T=0.7 and 25% at T=1 (Qwen2.5-7B: 1% and 21%), against at most 0.3% for the post-trained models. Invalid samples count as wrong, which deflates the base model at large k but inflates the post-trained model's gain at small k. Among valid samples only, the post-trained models still gain +12.0 [+9.8, +14.1] (Qwen3, T=0.7), +18.7 [+16.6, +20.7] (Qwen3, T=1), +12.2 [+10.4, +14.0] and +16.8 [+15.0, +18.5] points (Qwen2.5) at k=1, so a quarter to a third of the raw gain at T=1 is format. On the cases where every sample of both models is valid, the large-k loss persists: −15.2 [−21.9, −8.6] at k=32 for Qwen3 (105 cases) and −10.2 [−12.7, −7.6] for Qwen2.5 (748 cases), both at T=0.7. (At T=1 almost no case has 32 valid base samples.) The reported large-k losses are thus lower bounds.

C.4. Which post-training stage narrows the differential?

The Qwen pairs compare a base model with the end of a post-training pipeline. OLMo 3 (Team Olmo et al., 2025) releases every stage of its 7B instruct pipeline from

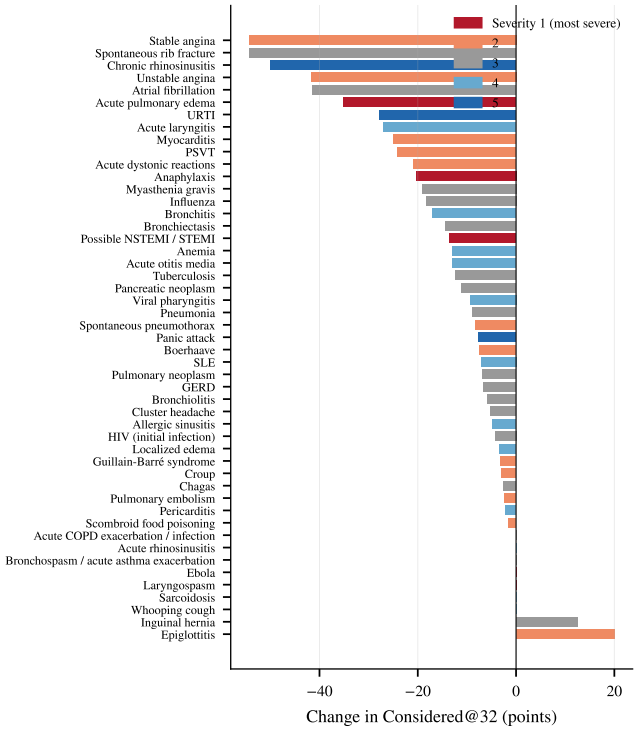


Figure 3: Change in Considered@32 per condition, Qwen2.5-7B-Instruct minus its base model (T=1, 980 cases), colored by DDXPlus severity (dark red = most severe). Conditions with fewer than five cases giving them reference probability ≥ 0.1 are omitted. Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops.

one base model: supervised fine-tuning, then DPO, then RL with verifiable rewards on math, code, instruction-following and chat prompts (the released OLMo-3-7B-Instruct). It also releases RL-Zero checkpoints trained with RL directly from the base; we use the general one, trained on chat prompts with an LM-judge reward. (The RL-Zero-Mix checkpoint does not load in our vLLM version.) Table 8 gives the results.

The coverage loss replicates in OLMo. At T=0.7, every post-trained stage is worse than the base at k=32: by -14.6 $[-17.5, -11.7]$ points after SFT, -9.2 $[-12.1, -6.1]$ after DPO, -28.3 $[-31.5, -25.0]$ after RLVR, and -13.3 $[-16.1, -10.5]$ for RL-Zero. Diversity falls at every stage of the pipeline, from 4.93 distinct answers for the base to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR.

Compared stage to stage, DPO recovers part of what SFT lost at large k ($+5.4$ $[+2.9, +8.0]$ points at k=32, and $+10.3$ $[+9.0, +11.6]$ at k=1) while narrowing the differential further (-0.71 distinct answers). The final RLVR stage is the largest single narrowing. Against the DPO model it loses -19.1 $[-21.8, -16.3]$ points at k=32, 0.92 distinct answers and 7.7 points of coverage@32, and it is also worse at k=1 (-5.9 $[-7.2, -4.6]$). It drops severe-tier Considered@32 to 0.25, a loss of -16.1 $[-18.6, -13.5]$ points relative to the base,

		pass@1		pass@32		Distinct	
Prompt	T	Base	Post	Base	Post	Base	Post
default	0.7	0.388	**0.558**	**0.730**	0.607	3.76	1.27
default	1.0	0.282	**0.556**	**0.789**	0.643	6.60	1.53
plain	0.7	0.352	**0.583**	**0.727**	0.655	3.72	1.28
plain	1.0	0.244	**0.586**	**0.792**	0.705	5.57	1.54
CoT	0.7	0.409	**0.652**	0.788	**0.835**	3.89	2.08
CoT	1.0	0.275	**0.642**	0.837	0.837	6.31	2.24

Table 7: Qwen3-8B-Base vs. Qwen3-8B on DDXPlus (980 cases, 32 samples) under three prompts. Bold marks the significantly better model (paired case bootstrap). Without reasoning, the base model wins at k=32; with CoT, the post-trained model matches or beats it, but its samples stay far less diverse.

against −7.4 for the DPO model. At T=1 the pattern is the same (RLVR vs. DPO: −12.7 [−15.4, −9.9] at k=32).

RL applied directly to the base narrows the differential less. At T=1, RL-Zero shows no detectable loss of the base model's pass@32 (−1.8 [−4.5, +0.8]; clustered [−5.8, +2.1]) and 6.28 of its 7.53 distinct answers, and it loses 4.0 [1.2, 6.9] points of severe-tier consideration, against 14.4 for the full instruct pipeline. At T=0.7 it still loses 13.3 points at k=32 (clustered [7.3, 19.7]), about as much as SFT.

The OLMo base model often breaks the output format: 40% of its samples are invalid at T=0.7 and 47% at T=1, while unparsable completions are 16–21% for the SFT model and at most 1.6% for the later stages. No case has 32 valid base samples, so the all-valid check of §C.3 is not possible here. Among valid samples, the post-trained models' gains at k=1 are small (−1.9 to +4.0 points at T=0.7, +1.3 to +8.1 at T=1), so most of OLMo's raw gain at k=1 is format. Invalid samples count as misses and are excluded from distinct answers and entropy, so the base model's coverage and diversity are understated and the losses above are conservative.

Across these pipelines, then, the collapse of the sampled differential is not a necessary effect of RL. It accumulates over the instruction-tuning pipeline; in OLMo 3 7B its largest step is RL applied on top of SFT and DPO, but at 32B it is SFT (§C.5), so which stage narrows most depends on the recipe. RL applied directly to a base model (OLMo RL-Zero at T=1, AlphaMed's base-initialized checkpoint, and our GRPO runs in §F) narrows the differential less, but not always: RL-Zero at T=0.7 loses about as much as SFT. On Phenopacket Store the instruct pipeline also loses coverage: every stage trails the base model at k=32 (−2.0 [−3.0, −1.1] points after SFT, −1.5 after DPO, −1.4 after RLVR; all significant), while RL-Zero does not (−0.6 [−1.4, +0.2]), from a base model at 4.5% (§E.2).

C.5. Replication at 32B

We repeat the DDXPlus comparisons at T=0.7 with Qwen2.5-32B and its Instruct model, and with every released stage of OLMo 3.1 32B, whose base is OLMo-3-1125-32B; there is no 32B RL-Zero checkpoint (Table 9). The sampled differential collapses at 32B too, by about

T	Stage	pass@1	pass@32	Distinct	Entropy	Cov@32	Severe C@32
0.7	Base	0.178	**0.705**	**4.93**	**1.16**	**0.351**	**0.412**
	SFT	0.222	0.559	4.01	0.87	0.257	0.373
	SFT+DPO	**0.324**	0.613	3.30	0.63	0.272	0.338
	SFT+DPO+RLVR (Instruct)	0.266	0.422	2.37	0.42	0.195	0.251
	RL-Zero (RL from base)	0.295	0.572	4.06	0.84	0.286	0.326
1.0	Base	0.128	**0.713**	**7.53**	**1.66**	**0.398**	**0.469**
	SFT	0.191	0.634	6.32	1.29	0.321	0.426
	SFT+DPO	**0.313**	0.688	4.48	0.87	0.312	0.410
	SFT+DPO+RLVR (Instruct)	0.272	0.561	3.58	0.68	0.259	0.325
	RL-Zero (RL from base)	0.279	0.695	6.28	1.21	0.367	0.429

Table 8: OLMo 3 7B post-training stages on DDXPlus (980 cases, 32 samples). Severe C@32 is severe-tier Considered@32 as in Table 6. Bold marks the best value in each block. Diversity falls at every stage of the instruct pipeline, and the RLVR stage loses the most coverage; RL from the base keeps the base model's pass@32 at T=1.

half as much. Post-training costs 4.9 [2.8, 7.1] points at pass@32 for Qwen2.5-32B and 4.0 [2.0, 5.9] for OLMo 3.1 32B (clustered $[-0.6, 10.2]$ and $[1.2, 6.8]$; neither survives correction), halves the number of distinct answers ($3.78 \to 1.68$ and $4.32 \to 2.15$), and lowers severe-tier consideration by 6.1 and 8.4 points. For Qwen the loss holds on the 421 cases where every base sample is valid (-4.8 $[-7.8, -1.7]$); the OLMo 32B base gives invalid answers on 27% of samples. With few-shot prompts, which nearly remove invalid answers, both losses are three to five times larger (§D.3). The stage attribution differs from 7B. In OLMo 3.1 32B the largest narrowing is SFT (pass@32 -12.1 $[-14.4, -9.9]$, clustered $[-18.0, -6.8]$, distinct answers -1.39, severe tier -11.9); DPO recovers most of it ($+10.9$ against SFT), and the final RL step is small (-2.8 $[-4.3, -1.1]$ at k=32, clustered $[-5.3, -0.5]$, -0.14 distinct answers). OLMo 3.1 32B and OLMo 3 7B were post-trained with different recipes, so the stage that narrows the samples most is a property of the pipeline, not of SFT, DPO or RL as such. Under this prompt the stated readout improves in both 32B families (§4.2; with few-shot prompts only Qwen2.5-32B's does robustly, §D.3), and voting stops paying off: majority voting over 32 samples gains the base models 7.0 and 8.4 points and the Instruct models 1.9 and 0.9 (the latter n.s.).

Appendix D. The stated readout: elicitation, robustness and audit

D.1. Robustness of the stated readout to elicitation

The stated readout above uses a five-item list. We repeat the Qwen pairs and OLMo 3 7B's base and Instruct models at T=0.7 with a prompt that asks for a probability on every one of the 49 conditions, summing to one, and take each sample's top answer to be its highest-probability condition (Table 10). The results match the short list. After post-training the stated distribution is closer to the reference (TV -0.052 for Qwen3-8B, -0.067 for Qwen2.5-7B-Instruct, -0.032 for OLMo-3-7B-Instruct) while the samples move away ($+0.012$ to $+0.038$). Qwen3-8B's stated probability again beats agreement over 32

Model	pass@1	pass@32	Distinct	Cov@32	Severe C@32	TV one list↓	TV samples↓	Vote@32 gain
Qwen2.5-32B (base)	0.500	**0.795**	**3.78**	**0.352**	**0.35**	0.726	**0.726**	+7.0
Qwen2.5-32B-Instruct	**0.604**	0.746	1.68	0.276	0.29	**0.674**	0.759	+1.9
OLMo-3-1125-32B (base)	0.326	**0.796**	**4.32**	**0.347**	**0.39**	0.746	**0.742**	+8.4
SFT	0.461	0.674	2.94	0.265	0.27	0.736	0.776	+3.6
SFT+DPO	**0.584**	0.784	2.29	0.293	0.32	**0.694**	0.755	+2.1
SFT+DPO+RLVR (Instruct)	0.560	0.756	2.15	0.283	0.30	0.701	0.763	+0.9

Table 9: 32B replication on DDXPlus (T=0.7, 980 cases, 32 samples). TV columns: distance to the reference differential of one stated list and of the sampled top-1 answers (§4.2). Vote@32 gain: accuracy of the majority answer over 32 samples minus single-call accuracy, in points (§4.3). Bold: best value per family.

	TV to reference↓		AUROC, single-call answer correct			Top-3: one
Model	One distribution	Samples	Stated p	Agree@5	Agree@32	dist. − 32 samples
Qwen3-8B-Base	0.733	**0.752**	0.647	0.809	**0.845**	−6.7
Qwen3-8B	**0.681**	0.790	**0.717**	0.609	0.647	+14.6
Qwen2.5-7B	0.769	**0.758**	0.688	0.792	**0.827**	−10.2
Qwen2.5-7B-Instruct	**0.702**	0.770	0.674	0.715	**0.748**	+1.6
OLMo-3-1025-7B	0.807	**0.766**	0.610	0.744	**0.786**	−15.3
OLMo-3-7B-Instruct	**0.775**	0.787	0.749	0.705	**0.759**	−9.5

Table 10: Full-distribution prompt on DDXPlus (T=0.7, 980 cases): a probability on every condition, top answer = highest stated probability. TV: one sample's stated distribution vs. the sampled top-1 answers (bold: better model of each pair). AUROC: bold marks each model's best signal.

samples (+0.069 [+0.033, +0.104]), Qwen2.5-7B-Instruct's again does not (−0.074), and OLMo-3-7B-Instruct's ties it (−0.010; +0.044 against five samples). The top three of one full distribution beats the three most frequent of 32 sampled answers for Qwen3-8B (+14.6 points) but not for the other two models.

D.2. Ruling out format, accuracy and floor effects

Table 12 tests simpler explanations of the stated-list result. *Format.* The general models' post-trained lists and base-model lists at T=0.7 are nearly all complete (81–99% name five on-list conditions with probabilities; Qwen base models at T=1, 61–62%; the medical models, 61–80%). Restricting to complete lists leaves the gaps unchanged (Qwen2.5-7B: 0.779 vs. 0.719). *Accuracy.* A more accurate model puts more stated mass on the true pathology, which the reference also favors. Within samples whose top answer is correct, and within those whose top answer is wrong, the post-trained Qwen list is still closer to the reference (−0.018 to −0.034 and −0.036 to −0.079 in TV across both temperatures; Table 12 shows T=0.7). *Floor.* One list can do no better than the reference's own top five, and a finite set of samples no better than as many draws from the reference. Measured as excess over these floors, post-training lowers the list's TV (Qwen3-8B 0.479 → 0.435) and raises the samples' (0.557 → 0.613). *True pathology removed.* With the true pathology

Pair (base → post)	Named less / more often (point estimates)	Significant (FDR 5%) less / more	Expected less often by chance	P(at least as many less)
Qwen3-8B, T=0.7	5 / 12	2 / 3	8.5	0.98
Qwen3-8B, T=1	6 / 11	2 / 4	8.5	0.94
Qwen2.5-7B, T=0.7	5 / 12	2 / 2	8.6	0.99
Qwen2.5-7B, T=1	4 / 13	2 / 5	8.6	0.99
Qwen2.5-32B, T=0.7	6 / 11	3 / 5	8.5	0.93
OLMo 3 7B, T=0.7	8 / 9	7 / 6	8.5	0.69
OLMo 3 7B, T=1	8 / 9	6 / 6	8.5	0.70
OLMo 3.1 32B, T=0.7	7 / 10	3 / 0	8.5	0.83

Table 11: The 17 severe conditions one by one (DDXPlus). For each condition, the share of a model's lists that name it, over the cases whose reference gives it any weight; differences post-trained minus base with intervals that resample the true pathologies (10,000 resamples) and Benjamini–Hochberg correction across the 17 conditions. The last two columns compare the number of conditions named less often with a null that swaps each case's base and post-trained lists at random (2,000 swaps).

removed from both list and reference and both renormalized, the post-trained Qwen and OLMo 3.1 32B lists are still closer to the reference (−0.018 to −0.056), and DDXPlus's set F1 against the reference's support improves by 0.040–0.083 (Table 20). TV, mass and DDF1 are not independent: a list's TV is the mass it leaves out plus the mass it misallocates among the conditions it names (Table 2), and the misallocation rises after post-training in seven of eight settings, so the TV gain comes from the named set. *Patient-blind list.* A fixed list of the five conditions with the highest average reference probability scores TV 0.794, mass 0.238 and DDF1 0.272. Every model's list beats it in TV except OLMo 3 7B's; in DDF1 several base lists do not (Qwen2.5-7B 0.245, OLMo 3 7B 0.225), and the post-trained Qwen and OLMo 3.1 32B lists exceed it by 0.035–0.061. *Over-naming.* As a share of named conditions, post-trained lists name fewer that the reference gives less than 1% (Qwen3-8B 0.59 → 0.55) and fewer unwarranted severe conditions (Qwen2.5-32B 0.74 → 0.65), but as a count per list the change is mixed: from 0.22 fewer unwarranted severe conditions (OLMo 3.1 32B) to 0.35 more (OLMo 3 7B). *Severe conditions one by one.* Table 11 tests each of the 17 severe conditions, counting it wherever the reference gives it any weight. After Benjamini–Hochberg correction, 2–7 conditions per pair are named significantly less often after post-training and 0–6 significantly more often, and no pair names more conditions less often than a random swap of base and post-trained lists would. Significant losses include acute pulmonary edema (Qwen3-8B 0.75 → 0.61, Qwen2.5-32B 0.63 → 0.44, OLMo 3 7B 0.56 → 0.31, OLMo 3.1 32B 0.57 → 0.41), stable angina (Qwen2.5-7B 0.25 → 0.08), NSTEMI/STEMI (OLMo 3 7B 0.31 → 0.14, OLMo 3.1 32B 0.49 → 0.36) and myocarditis (Qwen2.5-32B 0.14 → 0.05); significant gains include pulmonary embolism (Qwen2.5-7B 0.29 → 0.56, OLMo 3 7B 0.20 → 0.65), NSTEMI/STEMI (Qwen2.5-7B 0.32 → 0.52) and unstable angina (Qwen3-8B 0.14 → 0.37). Among single stages, OLMo 3 7B's RLVR step and OLMo 3.1 32B's SFT and RLVR steps do name more conditions less often than chance (12–13 of 17, p=0.03–0.04).

Model (T=0.7)	Complete lists	TV of one list↓ complete only	top-1 right	top-1 wrong	Excess TV over floor↓ One list	Samples	Severe named ref. < 0.1	pass@32 list-sampled
Qwen3-8B-Base	0.90	0.743	0.663	0.806	0.479	**0.557**	0.752	0.780
Qwen3-8B	0.99	**0.701**	**0.646**	**0.769**	**0.435**	0.613	**0.680**	**0.811**
Qwen2.5-7B	0.95	0.779	0.689	0.828	0.514	**0.575**	0.823	0.772
Qwen2.5-7B-Instruct	0.91	**0.719**	**0.655**	**0.773**	**0.452**	0.605	**0.796**	**0.802**
Qwen2.5-32B	0.96	0.725	0.661	0.795	0.462	**0.543**	0.735	0.850
Qwen2.5-32B-Instruct	0.98	**0.674**	**0.639**	**0.726**	**0.409**	0.579	**0.645**	**0.880**
OLMo-3-1025-7B	0.81	**0.794**	0.687	**0.837**	**0.524**	**0.522**	**0.747**	**0.719**
OLMo-3-7B-Instruct	0.95	0.823	**0.659**	0.882	0.558	0.671	0.779	0.577
OLMo-3-1125-32B	0.89	0.744	0.662	0.812	0.481	**0.532**	0.706	0.839
OLMo-3.1-32B-Instruct	0.99	**0.701**	**0.644**	**0.774**	**0.436**	0.583	**0.634**	**0.853**

Table 12: Robustness of the stated-list result (DDXPlus, T=0.7, 980 cases, valid samples). Complete lists: share naming five on-list conditions with valid probabilities. TV of one list on complete lists only, and on samples whose top answer is right or wrong. Excess TV: TV minus the expected TV of the matching oracle (the reference's top five for one list; as many reference draws as the model has valid samples for the samples). Severe named, ref. < 0.1: share of named severe conditions that the reference gives less than 0.1 (over-naming). pass@32 list-sampled: top answers drawn from each sample's stated list. Bold: better model of each pair. Excess TV is averaged over samples against each case's floor, so it is not exactly the difference of the mean TV and the mean floor. A uniform guesser scores about 0.48 at pass@32.

Sampling from the list. Replacing each sample's top answer with a draw from its own stated list raises pass@32 for every model. Compared like for like, post-trained lists reach higher than their base model's lists in five of seven comparisons (Qwen3-8B 0.811 vs. 0.780 at T=0.7, Qwen2.5-32B-Instruct 0.880 vs. 0.850, OLMo 3.1 32B Instruct 0.853 vs. 0.839), but not for Qwen3-8B at T=1 (0.820 vs. 0.832) or OLMo 3 7B (0.577 vs. 0.719), and a uniform guesser reaches 0.48. OLMo 3 7B remains the exception throughout: its Instruct list is closer to the reference when its top answer is correct but much farther when it is wrong.

D.3. Few-shot elicitation

The main analyses give every model the same zero-shot chat prompt, which base models were not trained to follow; they give invalid answers on up to 47% of samples, which we count as wrong. We therefore re-ran the five final base/post-trained pairs at T=0.7 with three worked examples in the prompt: three fixed DDXPlus training patients with different true pathologies, each followed by the top five conditions of its reference differential, with probabilities, in the JSON format the instructions ask for. Base models see the instructions and examples as a plain document and complete the next one (generation stops at the end of the list); post-trained models see the same instructions and examples as earlier chat turns. Invalid samples fall below 0.5% for every model (Table 13).

The examples help base models most. Their pass@1 rises from 0.178 to 0.306 for OLMo-3-1025-7B, from 0.326 to 0.528 for OLMo-3-1125-32B and from 0.500 to 0.647 for Qwen2.5-32B, and their pass@32 reaches 0.76–0.90, while their stated lists barely change (TV 0.712–

Model	Invalid (%)	pass@1	pass@32	Cov@32	TV, one list all	TV, one list w/o true	DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5
Qwen3-8B-Base	0.1	0.403	**0.809**	**0.383**	0.768	0.860	0.245	**0.732**	+8.4	−16.0	0.722	0.828
Qwen3-8B	0.0	**0.601**	0.639	0.221	**0.682**	**0.805**	**0.327**	0.786	+0.2	+14.5	0.693	0.563
Qwen2.5-7B (base)	0.4	0.398	**0.761**	**0.333**	0.784	0.879	0.238	**0.756**	+6.2	−15.6	0.734	0.792
Qwen2.5-7B-Instruct	0.0	**0.450**	0.535	0.220	**0.702**	**0.806**	**0.309**	0.798	+0.6	+16.7	0.757	0.594
Qwen2.5-32B (base)	0.1	0.647	**0.896**	**0.343**	0.712	0.847	0.298	**0.715**	+7.1	−8.2	0.726	0.838
Qwen2.5-32B-Instruct	0.0	**0.663**	0.744	0.252	**0.654**	**0.775**	**0.349**	0.758	+0.6	+10.1	0.718	0.647
OLMo-3-1025-7B (base)	0.2	0.306	**0.857**	**0.454**	**0.787**	**0.859**	**0.245**	**0.712**	+10.9	−22.0	0.674	0.823
OLMo-3-7B-Instruct	0.1	**0.338**	0.558	0.213	0.808	0.884	0.238	0.830	+0.7	+0.2	0.733	0.668
OLMo-3-1125-32B (base)	0.4	0.528	**0.888**	**0.397**	0.719	0.818	0.302	**0.699**	+8.9	−13.4	0.670	0.826
OLMo-3.1-32B-Instruct	0.0	**0.575**	0.697	0.249	**0.697**	**0.812**	**0.311**	0.772	+1.1	+3.7	0.772	0.684

Table 13: Few-shot elicitation on DDXPlus (T=0.7, 980 cases, 32 samples; three worked examples in the prompt for every model, §D.3). Invalid: samples naming no listed condition. TV: distance to the reference differential, for one stated list (with the true pathology removed from both in "w/o true") and for the sampled top answers. Vote@32 gain and top-3 list minus top-3 of the samples are in accuracy points. AUROC: for the single-call answer being correct, of its stated probability and of agreement with the next 4 samples. Bold: better of the pair.

0.787, against 0.726–0.794 zero-shot). The losses in the samples therefore grow (Table 14): post-training costs 15.2–29.9 points of pass@32 in every pair, including both 32B pairs, whose zero-shot losses did not survive correction, and every post-trained model's Cov@32 (0.21–0.25) is below every base model's (0.33–0.45) and below a uniform guesser's.

For the three Qwen pairs the stated-list improvement persists and mostly grows: TV −0.057 to −0.087, −0.055 to −0.073 with the true pathology removed, and DDF1 +0.051 to +0.081, all surviving correction. OLMo 3 7B remains the exception (TV +0.021, n.s.; +0.024 without the true pathology). For OLMo 3.1 32B the improvement shrinks: the post-trained list still places more reference mass on the conditions it names (+0.047), but its TV (−0.022), TV without the true pathology (−0.006) and DDF1 (+0.009) no longer survive correction. At 32B the stated-list result is therefore robust for Qwen2.5-32B but depends on the prompt for OLMo 3.1 32B. Post-trained lists still name 5–10 of the 17 severe conditions less often than their base model's.

The consequences replicate in all five pairs. Majority voting over 32 samples gains base models 6.2–10.9 points and post-trained models 0.2–1.1. The three most frequent of 32 sampled answers beat the top three of one stated list by 8.2–22.0 points for base models, while for post-trained models the list is ahead by 0.2–16.7. Agreement with four more samples is the better error signal for every base model (AUROC 0.79–0.84, against 0.67–0.73 for the stated probability), and the stated probability for every post-trained model (0.69–0.77, against 0.56–0.68). This check uses one temperature and one set of examples, and the examples show the reference's own lists, which may help every model match the reference.

D.4. A third family: Mistral-7B v0.3

We repeat the comparison with Mistral-7B-v0.3 and Mistral-7B-Instruct-v0.3, a family also studied by Jang et al. (2026). With the zero-shot prompt the base model does not answer:

Pair (base → post)	pass@32 (points)	TV samples	TV, one list all	TV, one list w/o true	DDF1	Reference mass of one list
Qwen3-8B	−12.4 / −17.0	+0.043 / +0.054	−0.043 / −0.087	−0.018† / −0.055	+0.042 / +0.081	+0.062 / +0.113
Qwen2.5-7B	−9.7 / −22.7	+0.028† / +0.042	−0.062 / −0.082	−0.041 / −0.073	+0.062 / +0.072	+0.073 / +0.124
Qwen2.5-32B	−4.9† / −15.2	+0.033 / +0.044	−0.053 / −0.057	−0.043 / −0.072	+0.051 / +0.051	+0.067 / +0.084
OLMo 3 7B	−28.3 / −29.9	+0.090 / +0.118	+0.029† / +0.021†	+0.034 / +0.024	−0.006† / −0.007†	−0.010† / −0.002†
OLMo 3.1 32B	−4.0† / −19.1	+0.020† / +0.074	−0.045 / −0.022†	−0.023 / −0.006†	+0.040 / +0.009†	+0.056 / +0.047

Table 14: Base → post-trained differences under the zero-shot prompt / the few-shot prompt (DDXPlus, T=0.7). †: not significant under the condition-clustered bootstrap with Holm correction (over 112 tests for the zero-shot and 70 for the few-shot pairs; Appendix G).

52% of its samples at T=0.7 and 76% at T=1 are invalid (it continues the document instead), and its valid answers mostly repeat the first listed condition, so its single-call accuracy is 6% and the zero-shot pair is not informative. With the few-shot prompt of §D.3 the base model answers validly (0.4% invalid), and the pair reproduces the other families (Table 15). Post-training raises pass@1 by 14.0 points and lowers pass@32 by 14.5, cuts distinct answers from 5.0 to 1.3 and Cov@32 by 0.138, and moves the samples away from the reference (TV +0.058). Voting gains the base model 5.6 points and the Instruct model 0.8; the three most frequent of 32 samples beat one list by 12.6 points for the base model, while for the Instruct model the list is ahead by 7.0; agreement is the better error signal for the base model (AUROC 0.826 against 0.671) and the stated probability for the Instruct model (0.747 against 0.613). The stated list improves modestly (TV −0.036 and DDF1 +0.046, surviving clustering and Holm correction over 42 tests; −0.023 with the true pathology removed, which does not), the smallest gain of the three families, and 11 of the 17 severe conditions are named less often. Mistral-Instruct also names 0.4 conditions per list that are not among the 49.

D.5. Thinking mode

Qwen3-8B can reason in a thinking block before it answers; the main runs turn this off. With thinking on (zero-shot prompt, T=0.7, up to 8,192 tokens; traces average about 1,500 tokens and all end before the limit), the same weights behave differently (Table 15). Accuracy rises at both ends: pass@1 is 0.671 against 0.559 without thinking, and pass@32 is 0.815, above both the non-thinking model (0.607) and the base model (0.732; +8.4 [1.7, 15.3] points, not surviving Holm correction over 42 tests). The samples are still narrow (1.76 distinct answers against 3.77 for the base model; Cov@32 0.283 against 0.323) but no farther from the reference than the base model's (TV 0.745 against 0.750). The stated list is the closest of any 7–8B model (TV 0.670, against 0.700 without thinking). The shift in error signals reverses: agreement is again the better signal (AUROC 0.693 with five samples, against 0.638 for the stated probability), as for base models, and voting gains 1.5 points (n.s.). The narrowing of the samples and the advantage of the stated probability are therefore properties of answering directly, which a reasoning mode largely undoes, consistent with the chain-of-thought prompt of §C.3.

Model	Invalid (%)	pass@1	pass@32	Distinct	Cov@32	TV, one list all	TV, one list w/o true	TV, one list DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5
Mistral-7B-v0.3 (base), few-shot	0.4	0.428	**0.791**	**5.02**	**0.355**	0.777	0.875	0.250	**0.741**	+5.6	−12.6	0.671	0.826
Mistral-7B-Instruct-v0.3, few-shot	0.3	**0.568**	0.646	1.34	0.217	**0.741**	**0.851**	**0.296**	0.799	+0.8	+7.0	0.747	0.613
Qwen3-8B-Base	11.3	0.388	0.732	**3.77**	**0.323**	0.743	0.833	0.276	0.750	+3.4	−7.3	0.595	0.772
Qwen3-8B, non-thinking	0.0	0.559	0.607	1.27	0.217	0.700	0.815	0.318	0.793	+0.3	+13.7	0.734	0.613
Qwen3-8B, thinking	0.0	**0.671**	**0.815**	1.76	0.283	**0.670**	**0.799**	**0.349**	**0.745**	+1.5	+3.0	0.638	0.693

Table 15: A third family and a reasoning mode on DDXPlus (T=0.7, 980 cases, 32 samples). Top: Mistral-7B v0.3 with the few-shot prompt of §D.3. Bottom: Qwen3-8B-Base and Qwen3-8B with thinking off (the main runs) and on, with the zero-shot prompt. Columns as in Table 13; Distinct: distinct top answers per 32 samples. Bold: best within each block.

D.6. Averaging stated lists across calls

Stated lists can be averaged across calls, as samples can be voted. For each case we average the first 5 or 32 valid stated lists, each renormalized over the conditions it names, and score the mean against the reference (Table 16). Averaging helps base models far more than post-trained ones, whose lists vary little between calls (with the zero-shot prompt, 32 averaged lists name 7.6–13.4 conditions, against 16.6–28.8 for base models). The mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700, Qwen2.5-7B 0.717 against 0.717, OLMo 3.1 32B 0.691 against 0.701). With five lists on both sides, the post-trained mean is closer for Qwen2.5-7B under both prompts (−0.025 and −0.027) and for Qwen2.5-32B and Qwen3-8B under one prompt each, not significantly closer for OLMo 3.1 32B (zero-shot) and Mistral (few-shot), and farther for OLMo 3 7B and few-shot OLMo 3.1 32B. Averaged base lists also name severe conditions at least as often, and more often for OLMo 3.1 32B, OLMo 3 7B, Mistral and few-shot Qwen2.5-32B (by 0.06–0.14). Direct-answer post-training thus mostly packs into one call the differential that the base model spreads across calls, as Lake et al. (2025) find for opinions, rather than adding information that averaged base lists lack. Qwen3-8B's thinking mode goes furthest: five of its lists beat five base-model lists by more than any direct-answer model's (TV −0.049) and name more severe conditions (+0.129). For a post-trained model answering directly, averaging five lists instead of reading one improves TV by only 0.005–0.025 at T=0.7 (up to 0.038 at T=1); it names 6–17 points more severe conditions, mostly by naming one to three more conditions. Five post-trained lists still cover 0.44–0.50 of the reference mass at T=0.7 (OLMo 3 7B 0.37) with six to eight conditions, about twice what their 32 samples cover (0.22–0.28) and as much as a uniform guesser naming about 24 conditions (0.48), so the coverage deficit of §4.1 is one of the sampled readout.

D.7. Which signal to use: combination and held-out audit

Whether the stated probability or agreement across samples is the better error signal varies across models, so we ask what predicts it and whether a practitioner can tell. We pool 56 model–setting combinations scored on DDXPlus: the base/post-trained pairs at both temperatures and with three prompts, the OLMo stages, the 32B models, the medical models, our GRPO runs, and the full-distribution prompt. They share weights, prompts and temperatures, so we treat the correlations below as descriptive. The stated probability's ad-

Pair	TV, base model 1 list	TV, base model mean of 5	TV, base model mean of 32	TV, post-trained model 1 list	TV, post-trained model mean of 5	TV, post-trained model mean of 32	Post − base, 5 lists each TV	Post − base, 5 lists each Severe named
Zero-shot prompt								
Qwen3-8B	0.743	0.699	0.691	0.700	0.695	0.692	−0.004†	+0.030†
Qwen2.5-7B	0.779	0.717	0.706	0.717	0.692	0.685	−0.025	+0.023†
Qwen2.5-32B	0.726	0.681	0.674	0.674	0.658	0.654	−0.024	−0.037†
OLMo 3 7B	0.794	0.730	0.717	0.823	0.805	0.801	+0.074	−0.076
OLMo 3.1 32B	0.746	0.691	0.678	0.701	0.677	0.673	−0.014†	−0.088
Few-shot prompt								
Qwen3-8B	0.768	0.704	0.689	0.682	0.673	0.672	−0.030	−0.006†
Qwen2.5-7B	0.784	0.712	0.694	0.702	0.686	0.683	−0.027	+0.012†
Qwen2.5-32B	0.712	0.655	0.641	0.654	0.640	0.638	−0.015†	−0.094
OLMo 3 7B	0.787	0.703	0.676	0.808	0.786	0.778	+0.083	−0.141
OLMo 3.1 32B	0.719	0.651	0.630	0.697	0.682	0.680	+0.031	−0.060
Mistral-7B v0.3	0.777	0.708	0.691	0.741	0.723	0.720	+0.015†	−0.118
Zero-shot prompt, thinking on								
Qwen3-8B, thinking	0.743	0.699	0.691	0.670	0.650	0.647	−0.049	+0.129

Table 16: Averaging stated lists across calls (DDXPlus, T=0.7, 980 cases). TV to the reference differential of one stated list (per-case mean over valid lists, as in Table 2) and of the mean of the first 5 or 32 valid lists, for the base and the final post-trained model. Last two columns: post-trained minus base with five averaged lists each, for TV and for the share of severe conditions with reference probability ≥ 0.1 that are named. †: the condition-clustered 95% interval includes zero.

vantage over five-sample agreement correlates with the *single-answer rate*, the share of cases whose five samples all give the same top answer (Spearman 0.75), with the stated probability's own AUROC (0.82; the gap contains this term, so part of this is by construction), and negatively with the number of distinct answers (−0.74). The stated probability is the better signal for 19 of the 56. No single-answer threshold separates them: Qwen2.5-7B-Instruct still favors agreement, while OLMo 3 7B's DPO and Instruct models favor their unusually informative stated probability. Combining the signals beats choosing between them. Fitting a logistic model of correctness on the stated probability and five-sample agreement with 100 labeled cases, and evaluating on the other 880, gives a mean held-out AUROC of 0.753 for the 39 post-trained combinations (0.765 for all 56); averaging the two signals' ranks without fitting gives 0.746; picking the better single signal on the same 100 cases gives 0.720; always using agreement 0.695; always using the stated probability 0.671. The logistic combination beats the single-signal pick for 35 of the 39 post-trained combinations. This matches the general finding that hybrids of consistency and verbalized confidence beat either (Xiong et al., 2024). For choosing a single signal, the simplest rule, always using agreement, is right for 66% of the combinations (54% of the 39 post-trained ones) and loses 0.025 AUROC on average (0.032). An audit does better: choose the signal with the higher AUROC on 100 random labeled cases, using five samples per case, and evaluate the choice on the other 880. Over 500 draws per combination it picks the better held-out signal 86% of the time (82% for post-trained models) and loses 0.005 AUROC (0.007); with 50 cases, 82% and 0.010; with 200, 90% and 0.003. It is reliable when the two signals differ by more

than 0.05 AUROC (96% of draws, 39 combinations) and close to a coin flip when they differ by less than 0.02 (53%, 5 combinations), where the choice matters least. The audit's main cost is gold labels for 100 cases from the target population, and we validate it only within DDXPlus.

Appendix E. Medical post-training and rare disease

E.1. Medical post-training

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed improves on its parent at every k (Figure 4): +3.9 points at k=1 ([+2.2, +5.5]; clustered [−0.4, +8.4]), widening to +8.4 at k=32 ([+6.0, +10.6]; clustered [+2.6, +14.7]). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.28 vs. 0.30), and coverage (0.27 vs. 0.24) stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier (+2.2 to +4.3 points; Table 6). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3% of samples, against 4.3% for its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution; the collapse comes from the general post-training stage. The other two descendants go further (Table 17). HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation from a reasoning teacher) both *widen* the differential relative to their parent: 2.49 and 3.29 distinct answers vs. 1.82, and coverage@32 of 0.31 and 0.33 vs. 0.24. Their pass@32 rises by 13.7 [+11.3, +16.0] and 18.1 [+15.5, +20.7] points (clustered [8.0, 20.0] and [10.5, 25.9]). m1 gains almost nothing at k=1 (+0.9, interval includes zero), so its improvement lies entirely in breadth. Both reasoning-trained models are also less calibrated (ECE 0.17 and 0.21 vs. 0.11). Medical post-training on an instruction-tuned model thus never adds to the collapse, and reasoning distillation partially reverses it.

Medical RL from the base model. Alongside the instruct-initialized AlphaMed-7B, the AlphaMed authors released a companion checkpoint, AlphaMed-7B-base-rl, initialized from Qwen2.5-7B rather than Qwen2.5-7B-Instruct (according to its configuration file). The paper (Liu et al., 2025) describes only the instruct-initialized models: GRPO with a binary correctness reward on about 19k multiple-choice questions from MedQA and MedMCQA, with a 64-question by 8-sample batch for 300 steps. It does not document the base-initialized checkpoint, so we cannot rule out differences in data or hyperparameters, and we treat it as corroborating rather than controlled evidence. It offers a third-party check on whether RL itself causes the collapse (Table 17, lower block). It does not. RL from the base model improves on the base at every k (+14.3 [+13.1, +15.7] points at k=1, +9.5 [+7.4, +11.6] at k=32). It keeps 3.14 distinct answers, against 1.87 for the instruct-initialized AlphaMed. Against that instruct-model AlphaMed, it is 2.2 [0.4, 3.9] points worse at k=1 and 10.8 [+8.6, +13.2] points better at k=32. Its severe-tier Considered@32 shows no detectable change relative to the base (−0.7 [−3.0, +1.7] points), against −12.4 for instruction tuning and −8.1 for instruction tuning followed by AlphaMed's RL (Table 6; all well below the chance levels of §C.2). RL from a base model also narrows less in OLMo's RL-Zero at T=1

Model	Post-training	pass@1	pass@32	Distinct	Entropy	Cov@32	ECE
Qwen2.5-7B-Instruct	(parent)	0.478	0.598	1.82	0.30	0.239	**0.106**
AlphaMed-7B	rule-based RL	0.516	0.682	1.87	0.28	0.266	0.121
HuatuoGPT-o1-7B	SFT + PPO	**0.536**	0.735	2.49	0.44	0.306	0.168
m1-7B-23K	SFT distillation	0.486	0.779	3.29	0.62	0.328	0.206
Qwen2.5-7B (base)	(parent)	0.352	0.695	**4.91**	**1.00**	**0.335**	0.291
AlphaMed-7B (base-RL)	rule-based RL	0.495	**0.790**	3.14	0.62	0.320	0.202

Table 17: Medical models on DDXPlus (980 cases, 32 samples, T=0.7). Upper block: descendants of Qwen2.5-7B-Instruct; none narrows the differential further, and the reasoning-trained models widen it. Lower block: AlphaMed's companion checkpoint trained from the base model, which keeps most of the base model's diversity and has the best zero-shot pass@32 of any 7B model.

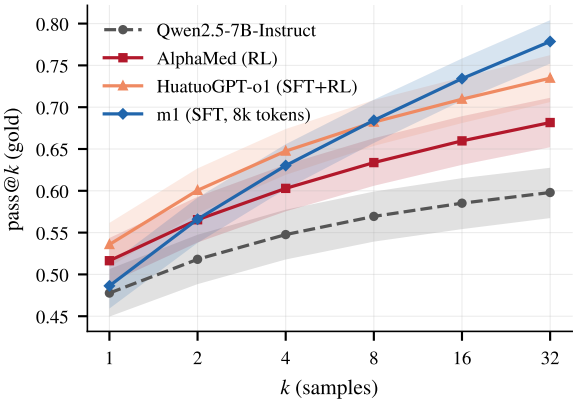


Figure 4: DDXPlus at T=0.7 (980 cases): Qwen2.5-7B-Instruct (dashed) vs. its medical descendants, with 95% case-bootstrap bands. Every descendant matches or beats its parent at every k, and the gaps widen with k: there is no sharpening tax at the medical stage.

and in our GRPO runs (§F), but not uniformly: RL-Zero at T=0.7 loses about as much as SFT, and within OLMo 3 7B's instruct pipeline the RL stage after SFT and DPO is the largest single narrowing (§C.4).

E.2. Rare disease: overconfidence without accuracy

On 1,401 Phenopacket Store cases covering 749 diseases, every 7–8B model is near the floor: pass@1 is 1.4–2.2% and pass@32 3.7–7.0% for the Qwen-family models, and lower for OLMo 3 7B (2.5–4.5%). Stated confidence is not near the floor. Qwen2.5-7B-Instruct states a mean top-1 probability of 0.51, and its RL-trained medical descendants state 0.67 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy is about 2%. ECE accordingly rises from 0.49 to 0.64–0.67; at this accuracy ECE mostly restates mean confidence, so we read it as a rise in confidence without a matching rise in accuracy. The RL-trained medical

Model	≤2022 (n=901) pass@1	pass@32	2023–24 (n=348) pass@1	pass@32	2025+ (n=152) pass@1	pass@32
Qwen2.5-7B (base)	2.0	8.2	0.0	0.3	0.6	3.3
Qwen2.5-7B-Instruct	2.3	7.7	0.0	0.9	0.8	3.9
AlphaMed-7B	3.0	9.0	0.3	1.4	1.4	3.9
AlphaMed-7B (base-RL)	2.4	8.8	0.1	1.1	1.4	4.6
HuatuoGPT-o1-7B	3.1	8.4	0.3	1.4	1.3	3.9
m1-7B	2.6	9.5	0.3	1.7	1.0	3.9
Qwen3-8B-Base	2.1	8.3	0.1	0.6	1.1	3.9
Qwen3-8B	3.1	5.2	0.1	0.3	2.0	2.6

Table 18: Phenopacket Store accuracy (%, exact diagnosis) by publication year of the source case report (1,401 cases, 32 samples, T=0.7). Every model is far more accurate on cases published up to 2022 than on 2023–24 cases; 2025+ cases fall in between.

models (AlphaMed; HuatuoGPT-o1 after SFT) thus state much higher confidence where they are least accurate. The effect also appears without instruction tuning: AlphaMed's base-initialized checkpoint is the most overconfident Qwen-family model (mean confidence 0.71, ECE 0.70, accuracy 1.7%). The medical descendants do gain a little accuracy: AlphaMed beats its parent at every k (+0.6 [+0.2, +0.9] points at k=1, +1.0 [+0.1, +2.1] at k=32), HuatuoGPT-o1 at $k \leq 8$, and m1 at $k \geq 2$ (+1.4 [+0.4, +2.5] at k=32). These gains are an order of magnitude smaller than the rise in confidence. m1, trained only with supervised reasoning traces (at its intended 8k-token budget), is the exception on calibration and diversity: it is the most diverse model (15.3 distinct answers per 32 samples, against 11.5 for the parent and 9.2 for HuatuoGPT-o1), and its confidence rises least (0.57).

General post-training reproduces the DDXPlus pattern on Qwen3. Qwen3-8B beats its base at pass@1 (+0.7 [+0.4, +1.1]) and trails it from k=8 on, by −2.2 [−3.1, −1.3] points at pass@32 (a relative loss of 37%), with half the distinct answers (6.4 vs. 13.4). For Qwen2.5 no pass@k difference is significant. Unlike medical RL, general post-training *lowers* stated confidence here (0.61 to 0.46 for Qwen3, 0.61 to 0.51 for Qwen2.5), although base models also give invalid answers more often (8.5% of Qwen3-8B-Base samples). OLMo 3 does not share this: every stage of its pipeline states a mean confidence of 0.73–0.84 at about 1% accuracy (pass@32 2.5–4.5%), and its SFT checkpoint is the most overconfident model we evaluate (confidence 0.84, ECE 0.83). On the 500-case subset, our 8B GRPO runs (§F) are indistinguishable from their base (all pass@k differences within ±0.6 points).

Accuracy depends strongly on when the source case report was published (Table 18). For every model, pass@32 is 5–10% on cases published up to 2022 (n=901), at most 1.7% on 2023–24 cases (n=348), and 2.6–4.6% on 2025+ cases (n=152). The pattern is not monotone, since 2025+ cases score above 2023–24 cases, so case mix also matters; the gap between older and recent cases is consistent with, but does not establish, memorization of older case reports.

Model	Single-call correct	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct			Correct answers
				Stated p	Agree@5	Agree@32	
Qwen3-8B-Base	0.074	+4.6	−7.0	0.567	0.768	**0.830**	104
Qwen3-8B	0.080	+0.4	+0.4	0.707	0.726	**0.763**	112
Qwen2.5-7B	0.062	+3.8	−5.4	0.598	0.788	**0.845**	87
Qwen2.5-7B-Instruct	0.056	+1.5	−3.6	0.717	0.661	**0.745**	78
OLMo-3-1025-7B	0.060	+3.1	−4.6	0.464	0.765	**0.814**	84
SFT	0.035	+1.1	−3.6	0.628	0.730	**0.814**	49
SFT+DPO	0.026	+1.2	−3.3	0.651	0.673	**0.725**	36
SFT+DPO+RLVR (Instruct)	0.035	+0.9	−2.7	0.652	0.644	**0.726**	49
RL-Zero (from base)	0.065	+2.3	−3.1	0.583	0.791	**0.827**	91
AlphaMed-7B	0.071	+1.6	−3.0	0.586	0.755	**0.797**	100
HuatuoGPT-o1-7B	0.093	+2.6	−5.7	0.653	0.708	**0.756**	130
m1-7B	0.084	+2.9	−7.9	0.650	0.747	**0.800**	117
AlphaMed-7B (base-RL)	0.054	+2.5	−4.2	0.672	0.733	**0.789**	75

Table 19: Aggregation and uncertainty on Phenopacket Store (1,401 real rare-disease cases, 32 samples, T=0.7). An answer counts as correct if it is the diagnosis or a closely related Mondo class. Columns as in Table 3; Correct answers: number of correct single-call answers, the positives for the AUROCs. Bold: best signal.

E.3. Stated and sampled readouts on real rare-disease cases

Phenopacket Store cases are real, but they have no reference differential and accuracy is near the floor, so we can test only the consequences of §4.3, and we count an answer as correct if it is the diagnosis or a closely related Mondo class (exact matches give only 8–31 correct single-call answers per model). Top answers are grouped by Mondo class for voting and agreement (Table 19). The DDXPlus shifts reappear, but the top-three comparison does not flip as it does on DDXPlus. Majority voting over 32 samples helps the base models by 3.1–4.6 points but the final post-trained models by only 0.4–1.5 (Qwen3-8B +0.4, n.s.). Post-training weakens agreement as an error signal (Qwen3-8B 0.830 → 0.763, Qwen2.5-7B 0.845 → 0.745, OLMo 3 7B 0.814 → 0.726) and strengthens the stated probability (0.567 → 0.707, 0.598 → 0.717, 0.464 → 0.652). Agreement beats the stated probability by 0.25–0.35 AUROC for the base models and by only 0.03–0.07 for the post-trained ones, significantly only for Qwen3-8B (-0.056 $[-0.120, -0.002]$). On these open-ended cases the three most frequent of 32 sampled answers still contain a correct diagnosis more often than one list's top three, but the gap shrinks from 4.6–7.0 points for the base models to 2.7–3.6 for Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct and to a tie for Qwen3-8B ($+0.4$ $[-0.8, +1.8]$). The medical models behave like their post-trained parent, with agreement the better signal for all of them.

Appendix F. GRPO with a group-level reward

We also fine-tuned Qwen3-1.7B (instruct) and Qwen3-8B-Base with GRPO (LoRA, 200 steps, 16 samples per prompt, KL coefficient 0.04) on DDXPlus training patients disjoint from the test split, comparing the standard correctness reward (`acc`) with a group-level reward (`gm`). If k_a of the n samples in a GRPO group give top answer a, `gm` adds $\text{clip}(n\, p_{\text{ref}}(a) - (k_a - 1), 0, 1)$, which pays an answer only while its condition is under-

represented relative to its reference mass. At 1.7B, a boosted variant (weight 3, training temperature 1.2, so the two changes are confounded) raised pass@1 by 1.6 and distinct answers by 0.10 over `acc` (two seeds; intervals ignore seed variation). From Qwen3-8B-Base, both recipes reached nearly the official Qwen3-8B's top-1 accuracy (0.537 and 0.539 vs. 0.558) while keeping more than twice its distinct answers (2.7–2.9 vs. 1.27), and `gm` narrowed the base model less than `acc` (distinct answers +0.14 at equal pass@1). Neither changed Phenopacket accuracy. Because `gm` needs a reference differential at training time, which real clinical data rarely provide, and the runs train on the evaluation distribution, these are existence results only. Closely related rewards computed from group answer counts have appeared since: a witness advantage that trains models to sample the distributions they state (Arabi et al., 2026), and a batch bonus that penalizes repeated answers (Song et al., 2025); we do not compare against them. The GRPO runs enter only the 56-combination audit (§D.7).

Appendix G. Condition-clustered intervals and multiple comparisons

DDXPlus has 20 test cases for each of 49 true pathologies, and effects can be specific to a condition, so cases are not independent. Table 20 gives the headline base-vs-post differences with a bootstrap that resamples the 49 pathologies with all their cases (10,000 resamples). Two-sided p-values use the add-one correction, $2(1+\min(n_{\leq 0}, n_{\geq 0}))/(B+1)$, so the smallest attainable value, 2/10,001, lies below the strictest Holm threshold, 0.05/112. Holm correction runs over the 112 tests (8 pairs $\times$ 14 metrics fixed in advance: pass@1, pass@32, distinct answers, coverage@32, sampled TV, list TV, mass named, severe named, list and sample TV without the true pathology, DDF1, unwarranted severe conditions named, and the two AUROCs; the table shows seven). The same procedure is applied separately to the few-shot pairs (70 tests), the Mistral and thinking-mode pairs (42) and the stage and medical comparisons (154). Not surviving correction among the headline tests, marked †: the 32B pass@32 losses, the sampled-TV increases for Qwen2.5-7B and OLMo 3.1 32B, OLMo 3 7B's list TV at both temperatures and its list TV without the true pathology at T=1, Qwen3-8B's list TV without the true pathology at T=0.7, OLMo 3 7B's DDF1 at T=0.7, and the drop in agreement AUROC for both OLMo models. OLMo 3 7B's list TV without the true pathology at T=0.7 survives as a significant *increase*.

Pair (post − base)	pass@32 (pts)	Distinct	TV samples	TV one list	TV list w/o gold	DDF1	AUROC agree@32
Qwen3-8B, T=0.7	−12.4	−2.50	+0.043	−0.043	−0.018†	+0.042	−0.160
	[−19.5, −5.9]	[−2.88, −2.13]	[+0.021, +0.063]	[−0.059, −0.028]	[−0.032, −0.005]	[+0.029, +0.056]	[−0.216, −0.097]
Qwen3-8B, T=1	−14.6	−5.06	+0.067	−0.074	−0.039	+0.066	−0.158
	[−21.6, −8.4]	[−5.49, −4.66]	[+0.042, +0.091]	[−0.090, −0.059]	[−0.052, −0.027]	[+0.053, +0.080]	[−0.211, −0.103]
Qwen2.5-7B,	−9.7	−3.09	+0.028†	−0.062	−0.041	+0.062	−0.106
T=0.7	[−16.2, −4.1]	[−3.61, −2.58]	[+0.007, +0.049]	[−0.076, −0.049]	[−0.054, −0.028]	[+0.049, +0.076]	[−0.162, −0.041]
Qwen2.5-7B, T=1	−10.3	−5.59	+0.042†	−0.088	−0.056	+0.083	−0.104
	[−17.4, −4.3]	[−6.04, −5.15]	[+0.015, +0.068]	[−0.103, −0.074]	[−0.069, −0.044]	[+0.069, +0.097]	[−0.160, −0.046]
Qwen2.5-32B,	−4.9†	−2.10	+0.033	−0.053	−0.043	+0.051	−0.079
T=0.7	[−10.2, +0.6]	[−2.50, −1.74]	[+0.018, +0.049]	[−0.064, −0.042]	[−0.053, −0.033]	[+0.041, +0.061]	[−0.119, −0.039]
OLMo 3 7B,	−28.3	−2.56	+0.090	+0.029†	+0.034	−0.006†	−0.076†
T=0.7	[−35.7, −21.2]	[−2.84, −2.27]	[+0.072, +0.109]	[+0.004, +0.052]	[+0.017, +0.051]	[−0.022, +0.009]	[−0.156, +0.000]
OLMo 3 7B, T=1	−15.2	−3.95	+0.079	−0.009†	+0.006†	+0.024	−0.076†
	[−20.9, −9.6]	[−4.22, −3.68]	[+0.060, +0.098]	[−0.030, +0.011]	[−0.006, +0.019]	[+0.010, +0.036]	[−0.150, −0.009]
OLMo 3.1 32B,	−4.0†	−2.18	+0.020†	−0.045	−0.023	+0.040	−0.056†
T=0.7	[−6.8, −1.2]	[−2.46, −1.89]	[+0.002, +0.037]	[−0.059, −0.031]	[−0.034, −0.012]	[+0.027, +0.053]	[−0.095, −0.018]

Table 20: Base → post-trained (final Instruct) differences on DDXPlus with 95% intervals from a bootstrap over the 49 true-pathology clusters. †: not significant after Holm correction across 112 tests (10,000 resamples, add-one p-values). pass@32 here is recomputed from the samples with the analysis scripts' name matching and can differ from Table 5 by up to 0.3 points.